Nonlinear 3D Euler: checked solver foundations

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Accuracy and optimality remain unproved. Lean has checked the named foundations listed below. The result is not yet a solver certified to reach a specified physical error, and no matching total-work lower bound has been established.

The physical problem

Full, unforced, compressible Euler with $\gamma = 2$ on the periodic cube $[0, 2\pi]^3$. The family has amplitude $A \in [0.5, 1]$ and density perturbation $\delta \in [-0.05, 0.05]$. Only initial conditions are supplied; all five conserved fields must evolve.

$\rho_0 = 1 + \delta \sin x \sin y \sin z$
$v_0 = A (\sin x \cos y \cos z, -\cos x \sin y \cos z, 0)$
$p_0 = 1 + (A^2/16)(\cos 2x + \cos 2y)(\cos 2z + 2)$
$E_0 = p_0 + \rho_0 |v_0|^2/2$

Initially $\rho_0 \geq 0.95$ and $p_0 \geq 0.625$. At $(0, 0, \pi/4)$, the vertical momentum equation forces $\partial_t(\rho v_z) = A^2/4 \geq 1/16$. This rules out freezing the initial zero vertical velocity. It does not prove that no analytic representation could ever exist.

Candidate and intended result

The current candidate is a conservative, explicit, global-speed Rusanov finite-volume method with exact-formula initial cell averages and a final-time-clipped CFL step. $T = 0.25$ is the numerical target; a common smooth existence interval is still unproved. The intended error includes the full evolved field, spatial reconstruction and time integration, using volume-normalized L1 error in each conserved component.

Required for completion	Current status
Common smooth time and uniform regularity	Unproved
Fully discrete nonlinear stability and continuum error	Unproved; foundations checked
Scientifically justified competitor class	Initial search scope; not finalized
Matching total-work upper and lower bounds	Unproved

All competitors receive the explicit initial-condition formulas. The target is not initial-data sampling complexity, a cache schedule, a CFL count, an output-size bound, or a ranking of a finite benchmark catalogue.

What Lean has checked

The retained audit checks **37 named statements** with Lean (version 4.24.0, x86_64-unknown-linux-gnu, commit 797c613eb9b6d4ec95db23e3e00af9ac6657f24b, Release). Transitive axiom dependencies are limited to the standard principles propext, Classical.choice and Quot.sound. Statements must still be interpreted with their definitions and explicit premises.

Layer	Checked claim and its limit
Actual nonlinear finite-volume update	Exact-real positive density/pressure from computed wave speeds and a valid CFL ratio; periodic conservation of all five components. This does not prove convergence.
Conservative / primitive connection	Actual derivative rules, invertible state-map derivative, equivalent Euler residuals, and a positive symmetrizing energy. This does not construct a solution.
Frozen principal update	The constant-coefficient linearized update contracts its discrete energy under the stated wave-speed and CFL bounds.
Varying principal update	An exact energy defect and an explicit growth estimate from stated spatial difference bounds. This is not the full nonlinear update stability theorem.
Grid embedding and state-box bounds	A mesh-independent 3D embedding and explicit coercivity/coefficient bounds. The trajectory's derivative energies and state-box membership are not bounded yet.
Actual mathematical entropy	Conserved-variable gradient and positive Hessian, convexity on admissible states, finite averaging bound, canonical entropy-flux potential and local entropy balance implied by differentiable Euler balance.
Actual finite-step entropy estimate	Flux residual, six-neighbor increment bound and finite-step Taylor remainder combine into entropy dissipation under explicit convex-region and step conditions. These conditions are not established for the full numerical trajectory.

The new stability estimate

Write $q = 1/\rho$, $e = (e_q, e_v, e_p)$ and $g_{q,p}(e,e) = e_p^2/(2p) + |e_v|^2/q + (e_q/q + e_p/(2p))^2$. Sum this over cells to obtain E. For the principal update, the proof gives
$\mathbf{E_{after} \le [1 + 756\,\Delta t\,(256\,B\,K + 32\,D)]\,E_{before}}$.

Both energies use the old local metric. Premises include $1/4 \le q,p \le 4$, $|a_d| + |v_d| \le B$, a valid CFL ratio, an identity central shift, and neighbor differences satisfying $|\Delta q| + |\Delta p| \le hK$ and $|\Delta v_d| \le hD$. Here $\Delta t = \text{ratio} \times h$. The constant does not depend on the number of cells when B, K and D have uniform bounds. Proving such bounds, the time variation of the metric and the nonlinear remainder are outstanding steps.

Exact negative control: with two cells, $q = (1,2)$, $p = 1$, $v = 0$, directional speeds 2 and ratio 1/12, the CFL number is 1/2. A specific error field has energy 5 before and $125/16 = 7.8125$ after the principal update. Thus a varying local metric need not contract. This error-field example does not replace the nonlinear Taylor–Green research family.

Grid control and numerical evidence

The checked embedding gives $|f(i,j,k)|^2 \le 8\, E_3(f)$. In one dimension, $E_1(f) = n^{-1} \Sigma f^2 + n \Sigma(f_{k+1} - f_k)^2$; E_2 and E_3 apply this construction recursively in the other directions. It contains the eight mixed differences with each coordinate differentiated at most once, scaled to the unit cube. The constant 8 is independent of n. This controls values from derivative energies; it does not bound those energies along the Euler evolution.

Independent candidate checks

Initial cell averages agree with independent eight-point-per-axis Gauss quadrature to 9.99e-16 in the tested cells. A randomized admissible-field update has conservation drift 3.55e-15. An intentionally too-small split speed gives pressure −49, independently reproduced by Lean. These checks do not certify floating-point error or continuum accuracy.

n	A / δ	Steps	min ρ	min p	max |v_z|
16	0.5 / -0.05	11	0.964880	0.971706	0.008675
32	0.5 / -0.05	23	0.957623	0.944254	0.012397
16	0.75 / 0	12	0.997845	0.944951	0.018992
32	0.75 / 0	25	0.997943	0.882108	0.027463
16	1 / 0.05	13	0.969251	0.916210	0.032740
32	1 / 0.05	27	0.961357	0.803544	0.047906

All six runs end at T = 0.25. They show evolving vertical velocity and admissible sampled states on two grids. Their success neither proves the full parameter family remains smooth nor supplies an exact continuum reference. No speedup or optimality claim is inferred from these runs.

Reproducibility and next proof obligations

The source is under demos/lean_euler/nonlinear/formal. record/solver contains the frozen assembled Lean input, generated audit, logs, result JSON, component hashes, numerical checks and local-lab job receipts. The Python method is not a verified extraction of the Lean definitions.

Run from the repository root with a new output directory:
.venv/bin/python demos/lean_euler/nonlinear/run_solver_study.py
--output artifacts/nonlinear-energy-replay

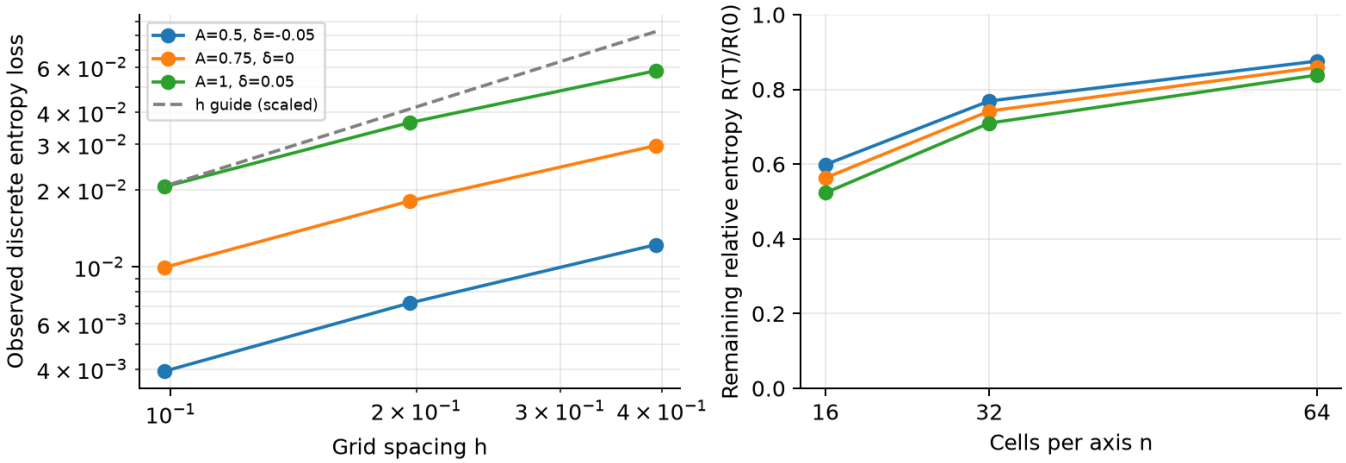
The next required bridge is quantitative regularity and a stability/error estimate for the actual nonlinear update. Initialization and continuous reconstruction must then be connected to the evolved PDE solution. A complete cost model and a matching necessary-accuracy lower bound remain essential before the goal can be completed.

Assembled proof SHA-256: 1c9be355e73eca7921f946968ffaf533
82fd2a719a489994327827d7407f8529
Draft PR: github.com/tomzhu0225/simjecture/pull/90

Entropy: the proposed error mechanism

For γ = 2, the mathematical entropy is η(U) = −ρ log(p/ρ²). Lean now checks its actual gradient and Hessian, convexity on the admissible conserved-state set, and the local entropy balance that follows from differentiable Euler balance. The Hessian was derived from η, not defined as a convenient positive form. Existence of a smooth solution and the periodic integral conservation theorem are still required.

A further checked result says a finite averaging map with nonnegative weights and unit row/column sums cannot increase this convex entropy. A bridge to the actual continuous output reconstruction remains unfinished.



Observed quantities, not certified continuum errors. Nine runs of the actual Rusanov update use n = 16, 32, 64, CFL = 0.3 and T = 0.25. Entropy loss is measured between discrete initial and final states. R is relative entropy about the conserved grid mean. The scaled h line is a visual guide; no rate or proof constant is inferred from this finite sample.

A / δ (n = 64)	Entropy loss / h	R(T) / R(0)	Min observed drop / (Δt h R)
0.5 / -0.05	0.039945	0.875728	5.368747
0.75 / 0	0.101323	0.859054	6.125759
1 / 0.05	0.208938	0.838228	7.076122

Remaining argument. Discharge the region and step conditions in the new finite-step theorem on the next page. Connect accumulated loss to final continuum error through proved state bounds and the reconstruction. Only then could a necessary mesh resolution support a total-work lower bound. The data above do not discharge any of these obligations.

The initial search scope includes higher-order methods; first-order Rusanov cannot be declared optimal in that scope without further argument. The proposed LF/Rusanov class with convex reconstruction is still under investigation, with explicit initial formulas and all preprocessing available to competitors.

Reference: Tadmor, Acta Numerica 12 (2003), §§7–8, on the balance between spatial entropy dissipation and explicit time discretization. The reference is not imported as a Lean theorem.

A checked finite-step theorem

The new theorem concerns the actual nonlinear Rusanov finite-volume update. It includes explicit time discretization. It is a conditional method theorem; the required uniform state region and the constants for the complete solve remain to be established.

The region conditions

Let K be a convex set of positive-density, positive-pressure conserved states. Both the old and the updated cell states must lie in K. Throughout K, for every conserved-state tangent k, require
$\lambda\,|k|^2 \le D^2\eta(U)[k,k] \le M\,|k|^2$,
and $|F_d(V) - F_d(U)|^2 \le L^2\,|V - U|^2$ for every direction. These are bounds on the actual physical entropy Hessian and flux, using the sum of the five squared nondimensional components. They are not numerical error estimates supplied as assumptions.

The viscosity and timestep conditions

Write $r = \Delta t/h$ and $0 < a_{min} \le a_d \le a_{max}$ for the directional global viscosities. A sufficient pair of conditions is
$M(1 + L^2) \le a_{min}\lambda/2$,
$3\,M\,r\,(a_{max}^2 + L^2) \le a_{min}\lambda/8$.

With $H_{total} = \Sigma_{cells}\,\eta(U)$ and $J = \Sigma_{forward\ periodic\ faces}\,|U_{right} - U_{left}|^2$, Lean proves
H_{total}(after) $\le$ H_{total}(before) $-$ (r a_{min} λ/8) J.

How the proof connects to the update

Checked piece	Role
Actual entropy calculus	The derivative of the actual entropy gradient pairs with the Euler flux to give the momentum component of the tangent.
Rusanov face residual	The Hessian and flux bounds control the central flux term; sufficiently large LF viscosity supplies a negative quadratic face contribution.
Periodic sum and finite step	The entropy potential telescopes. A six-neighbor quadratic bound and an actual Hessian-based Taylor estimate control temporal entropy production.

Practical limitation. These sufficient conditions may require a stronger viscosity or smaller step than the current CFL = 0.3 numerical runs. Those runs have not been certified to satisfy them. No uniform region, continuum error, cost lower bound or optimality result follows until the outstanding hypotheses and the full-solve bridges are discharged.